

BONAVISTA, NL, Canada

WMO: 711780

Lat: 48.670N

Lon: 53.110W

Elev: 84

StdP: 14.65

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.9	11.1	-0.3	5.4	9.3	2.9	6.4	12.4	57.1	25.0	53.3	25.6	25.4	290	1.003

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	75.3	66.0	72.5	64.3	69.9	63.0	67.8	73.3	66.0	70.7	64.2	68.5	16.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	95.1	70.8	63.9	89.2	68.8	62.1	83.7	67.2	32.2	73.7	30.8	70.9	29.4	68.5	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 49.1	43.1	39.5	4.5	81.1	4.8	1.8	1.0	82.4	-1.8	83.5	-4.5	84.5	-8.0	85.9		
(5)			3.0	70.9	5.0	1.9	-0.6	72.3	-3.6	73.4	-6.4	74.5	-10.1	75.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	25.8	24.4	27.9	34.8	41.7	49.9	60.3	61.7	55.1	46.4	38.7	30.7	(6)	(7)
	DBStd	14.14	6.84	7.38	6.50	5.29	6.29	6.98	6.53	5.11	5.66	5.67	5.87	6.43	(7)	(8)
	HDD50	4075	751	717	686	457	271	89	6	0	16	144	341	597	(8)	(9)
	HDD65	8611	1216	1137	1151	906	721	454	169	128	301	578	788	1062	(9)	(10)
	CDD50	990	0	0	0	0	14	86	327	362	168	31	2	0	(10)	(11)
	CDD65	53	0	0	0	0	0	1	25	24	3	0	0	0	(11)	(12)
	CDH74	133	0	0	0	0	0	2	77	50	4	0	0	0	(12)	(13)
	CDH80	6	0	0	0	0	0	0	4	2	0	0	0	0	(13)	(14)
(14) Wind	WSAvg	19.6	24.7	23.3	21.3	18.8	16.4	15.8	15.2	15.1	18.0	20.5	22.1	24.2	(14)	(15)
(15) Precipitation	PrecAvg	42.0	4.1	3.9	3.7	3.1	2.9	3.0	2.6	3.2	3.5	4.2	4.1	4.1	(15)	(16)
	PrecMax	51.1	7.0	8.0	6.8	5.7	5.0	7.4	6.5	7.0	6.4	7.1	7.4	7.2	(16)	(17)
	PrecMin	32.0	1.5	1.3	2.0	0.8	0.8	0.8	1.1	0.6	1.2	2.0	2.0	1.6	(17)	(18)
	PrecStd	5.9	1.4	1.8	1.3	1.2	1.1	1.5	1.2	1.5	1.4	1.4	1.4	1.2	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	47.0	47.5	55.8	66.8	72.9	79.7	78.6	73.1	64.7	57.9	52.1	(19)	(20)
		MCWB	48.1	45.8	45.2	48.9	56.7	62.3	68.3	68.1	65.6	61.0	55.5	50.6	(20)	(21)
	2%	DB	42.6	40.6	42.4	50.3	61.2	69.1	76.1	75.0	69.1	61.0	54.4	46.4	(21)	(22)
		MCWB	41.5	39.6	39.6	45.8	53.6	60.0	66.0	66.2	63.2	57.6	52.7	45.4	(22)	(23)
	5%	DB	37.8	36.8	38.3	45.8	56.5	65.4	73.2	72.4	66.4	57.8	50.9	42.3	(23)	(24)
		MCWB	36.7	35.6	36.1	42.0	50.8	57.9	64.6	64.6	61.5	54.7	49.2	41.1	(24)	(25)
	10%	DB	34.8	33.9	35.5	42.3	52.4	61.5	70.3	69.7	63.8	54.8	47.8	39.1	(25)	(26)
		MCWB	33.6	32.6	33.5	39.1	47.7	55.8	63.3	63.2	59.3	51.8	46.1	38.0	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	45.7	45.7	51.3	58.4	64.8	70.5	70.7	67.6	62.5	55.8	51.0	(27)	(28)
		MCDB	49.4	46.8	47.0	54.0	63.7	70.8	77.0	75.6	71.2	64.5	57.1	51.9	(28)	(29)
	2%	WB	41.6	39.8	40.0	46.5	54.9	61.6	68.0	67.9	64.6	58.4	52.8	45.5	(29)	(30)
		MCDB	42.5	40.5	41.7	49.7	59.4	66.6	74.2	73.3	67.9	60.2	54.0	46.3	(30)	(31)
	5%	WB	36.8	35.9	36.4	42.7	51.6	58.7	66.2	66.0	62.2	55.2	49.3	41.4	(31)	(32)
		MCDB	37.5	36.6	37.9	45.6	55.9	63.8	71.1	70.7	65.5	57.5	50.4	42.1	(32)	(33)
	10%	WB	33.8	32.8	33.9	39.4	48.0	56.0	64.3	64.2	60.0	52.2	46.0	38.2	(33)	(34)
		MCDB	34.6	33.5	35.2	41.8	52.5	60.9	68.9	68.4	63.1	54.3	47.4	39.0	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.0	9.3	9.8	12.2	13.7	14.5	12.1	11.0	8.7	8.7	8.3	(35)	(36)
		MCDBR	15.8	14.5	13.5	16.5	21.1	21.4	18.8	16.1	14.5	13.1	13.8	13.9	(36)	(37)
		MCWBR	16.6	14.6	12.2	12.7	14.4	13.8	10.9	9.6	9.9	11.4	13.9	14.6	(37)	(38)
	5% WB	MCDBR	15.6	14.2	13.1	16.5	19.8	20.0	17.0	15.0	14.0	13.2	13.7	13.9	(38)	(39)
		MCWBR	16.4	14.6	12.5	13.5	14.6	14.0	11.2	10.0	10.7	12.2	14.1	14.6	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.310	0.304	0.336	0.367	0.374	0.401	0.422	0.398	0.371	0.349	0.322	0.309	(40)	(41)
	taud		2.311	2.383	2.295	2.330	2.408	2.360	2.348	2.420	2.469	2.471	2.452	2.330	(41)	(42)
	Ebn at Noon		235	270	277	279	281	273	265	267	264	251	233	219	(42)	(43)
	Edh at Noon		23	27	34	37	36	38	38	34	29	25	20	20	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		290	539	893	1276	1517	1632	1662	1398	1076	635	353	226	(44)	(45)
	RadStd		25	48	92	128	124	131	156	95	74	70	44	21	(45)	(46)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page